Bangalore Institute of Technology

An Autonomous Institution Under VTU
K.R Road, V.V Puram, Bangalore – 560004

Board of Computer Science (CS&E /IS&E /AI&ML /CSE-DS /CSE-ICB)

AI MODULEWISE QUESTION BANK -2026-27 (ODD)
Batch: 2024

Course Title : Artificial Intelligence | **Course Code :BCS15B**
No. of Lecture Hrs./Week : 3:0:0:0 | **Exam Hours : 03**
CIE Marks:50 | **SEE Marks : 50**

Module-2 Questions.

1. What is well defined problem and solutions? Give pseudo code of simple problem solving agent.
2. Explain the following terms with an example:
   a. Initial state
   b. Goal state
   c. Actions
   d. Transition model
   e. Path cost
3. Explain the following problem with example
   a) Sliding-Block Puzzles
   b) Traveling Salesperson Problem
   c) Robot Navigation
4. Describe an informal description of the general tree-search and graph-search algorithms with pseudocode.
5. Summarize performance measurements of problem-solutions are performed.
6. a) Using Figure 2.1, formulate the route-finding problem from Arad to Bucharest. Identify the initial state, goal test, actions and path cost.
   b) Using Figure 2.1, give one possible solution path and calculate its path cost using the shown edge costs.
   c) Using Figure 2.1, identify an example of a repeated state or redundant path that could arise during search.

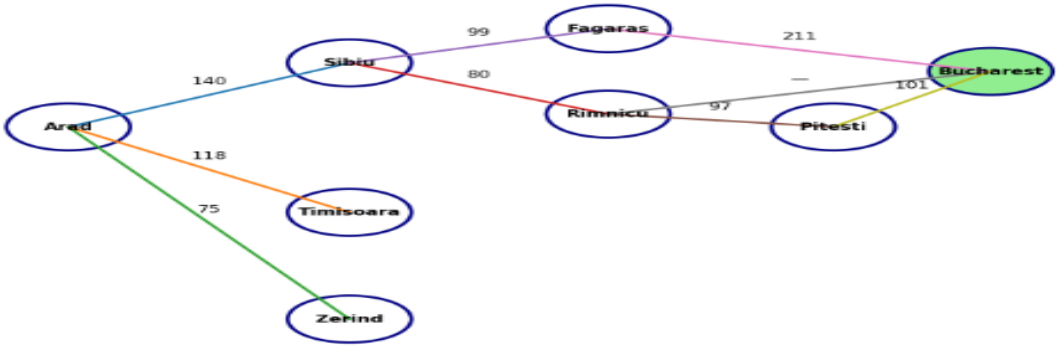


Figure 2.1

7. Discuss the vacuum world, 8-puzzle, and 8-queens as toy problems. Compare their state representations and action models
8. Explain how the incremental formulation can reduce the search space by prohibiting placement of a queen in an attacked square.
9. Explain breadth-first search with algorithm, data structure, completeness, optimality condition, time complexity, and space complexity.
10. Differentiate between Breadth-First Search (BFS) and Depth-First Search (DFS)
11. Explain the followings searching techniques.
    a) Uniform-cost search
    b) Depth-limited search
    c) Iterative deepening depth-first search
    d) Bidirectional search
12. Explain depth-first search with its data structure, completeness issues, optimality, time complexity, space complexity, and backtracking variant.
13. Determine the uniform least cost path from S to G and calculate its total path cost for the figure 2.2

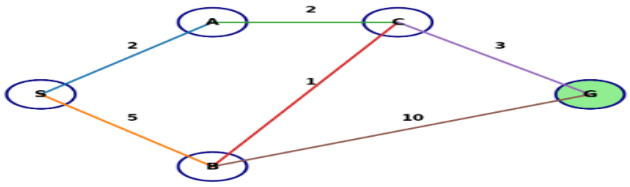


Figure 2.2

14. Determine the uniform least cost path from S to G and calculate its total path cost for the figure 2.3

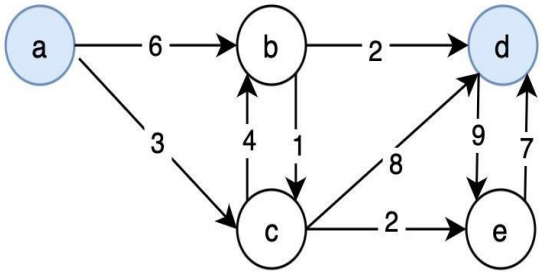


Figure -2.3: Here Initial Node =a and Goal Node = d

15. Explain the working of Depth-Limited Search with the help of an algorithm.
16. Explain the working of Iterative Deepening Search with a suitable search tree.
17. Explain Bidirectional Search with a suitable diagram.

18. Compare BFS, UCS and DFS based on completeness, optimality, time complexity and space complexity.

19. Consider the following search tree figure -2.4. Apply Iterative Deepening Search with depth limits 0, 1, 2 and 3. Show the nodes explored in each iteration and identify the iteration in which the goal is found.

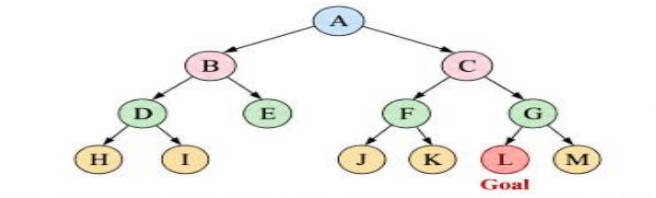


Figure 2.4

20. Consider the following problem search trees. Apply BFS, DFS, Depth limit, and Iterative Deepening Search with depth limits 0, 1, 2 and 3.

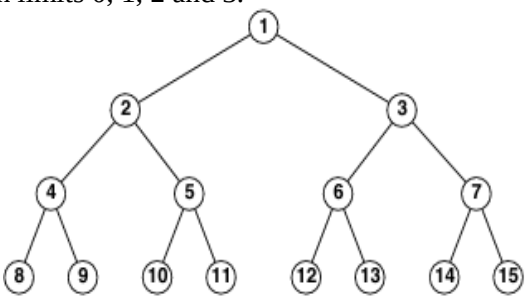


Figure-2.5: Here Initial Node=1 and Goal Node =11

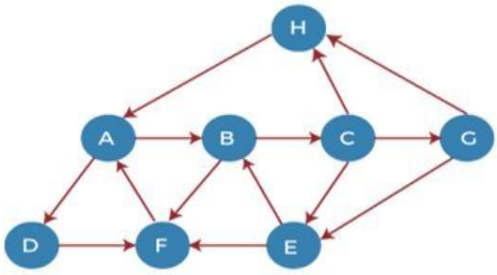


Figure-2.6:Here Initial Node=H and Goal Nodes= F and also E

Faculty In-charges

Course Coordinator **Module Coordinator** **IQAC** **Program Coordinator**